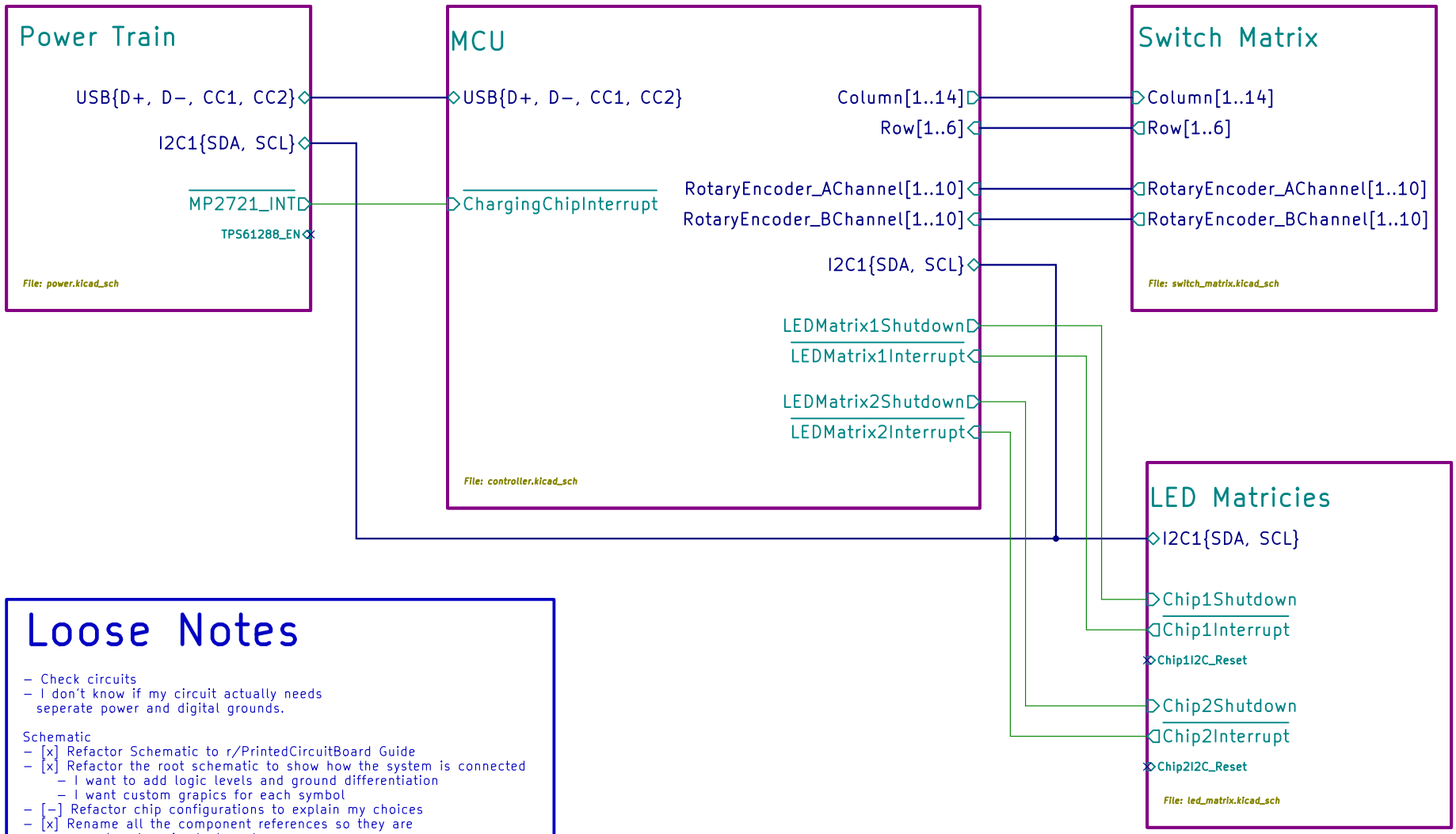


Loose Notes

- Check circuits
- I don't know if my circuit actually needs seperate power and digital grounds.

Schematic
- [x] Refactor Schematic to r/PrintedCircuitBoard Guide
- [x] Refactor the root schematic to show how the system is connected
  - I want to add logic levels and ground differentiation
  - I want custom grapics for each symbol
- [−] Refactor chip configurations to explain my choices
- [x] Rename all the component references so they are grouped and easier to layout
- [x] Select actual components, footprints, 3D models for all components
- [x] Pass those design reviews softwares
- [x] take review advice
- [ ] remove the outer switches from the thumb clusters

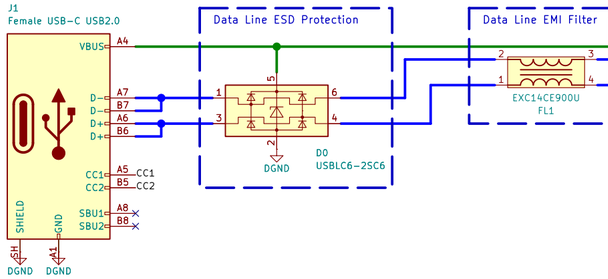
KiCad Feature Requests:
Hierarchical Bus
Hierarchical Sheet Symbols

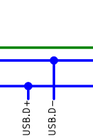
Sheet: /
File: slack.kicad_sch

Title: Slack

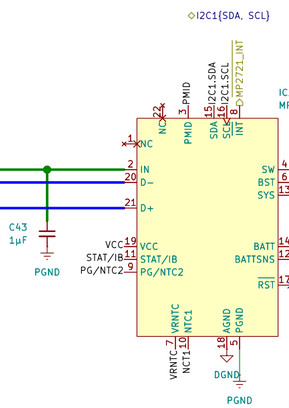
Size: A	Date: 2026-07-25	**Rev: A.0.7**
KiCad E.D.A. 10.0.5		Id: 1/7

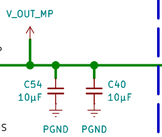
USB-C 2.0

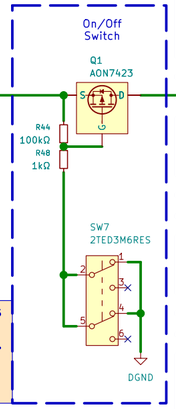




Charger

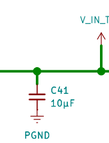


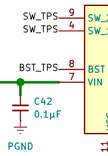


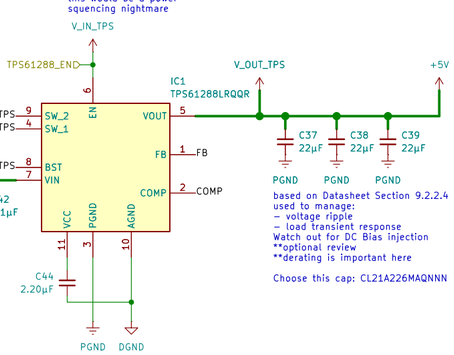


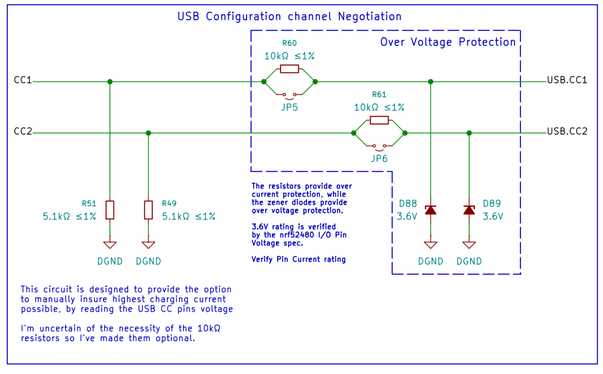
Few desirable switches are rated for max current of this system. So we're using a load switch MOSFET.

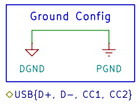
Boost Converter

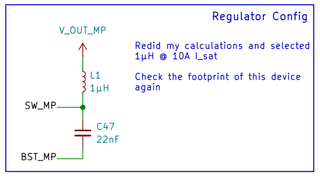


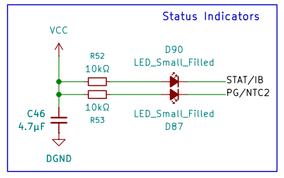


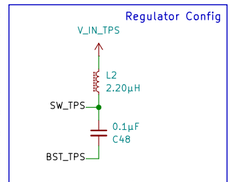


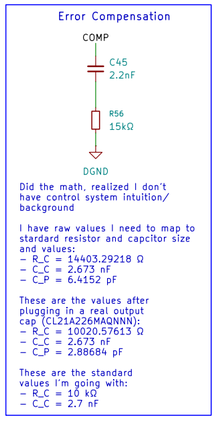


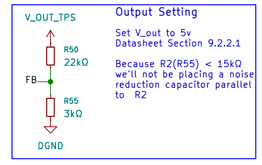


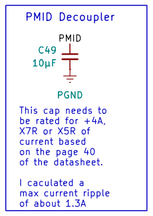


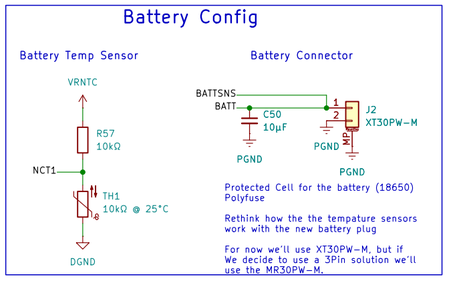












Do I have enough protect on the USB contector?
Do I still need to add resistors or pull up these lines?
Have I properly set up the charger and regulator correctly?
why can charge current be higher than input current in battery charger
Zener Diodes vs Schottky Diode for pin protection

Power Ratings per Critical Line
USB-C plug → Charger IC input:
– max 5V @ 3A
– must be configured

USB-C plug → Charger IC output:
– Worst case output current draw 6A
– Max output current 5A
– Voltage range 3.125V ↔ 3.913V
default 3.7V

Battery → Charger IC output:
– Worst case discharge current draw 7.5A
– Max discharge current 8A
– Battery Voltage 3V ↔ 4.2V

Voltage Regulator: Rating 5A, 5.5V
– max 5V @ 4A

FIRMWARE NOTE:
Power scenarios must be accounted for in the firware.

Selection of passives:
resistors, capcitors, inductors, diodes, switchs, LEDs, usb port

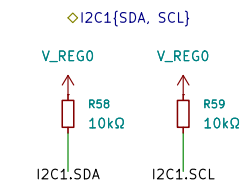
Consider BQ25640 & BQ25895

Sheet: /Power Train/	
File: power.kicad_sch	
Title: Slack	
Size: User	Date: 2026-07-25
KiCad E.D.A. 10.0.5	Id: 2/7

Rev: A.0.7

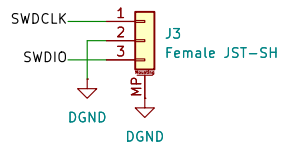
Microcontroller

I2C Bus Config



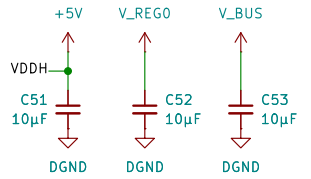
pick theses last based on the capacitance of the line on the PCB

Programmer



Verify that this is enough to program this board

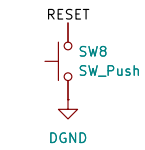
Power Config



Regulator 0 & Regulator 1 are both set to LDO mode

VDD is REG0, it defaults to +1V8 and can be set to V_REG0.

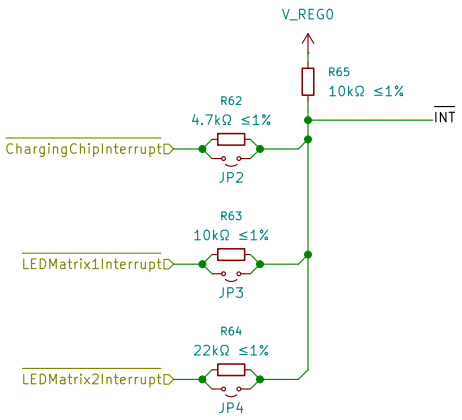
Reset Button



The datasheet in unclear about behavior of this pin

I't is possible we may need to set this manually

Interrupt Multiplexing



This circuit is designed to allow for both a analog and digital method of interrupt detection.

It uses a voltage ladder.
All values varified

Active interrupts	Node voltage
none	3.30 V
driver #2 (22 k)	2.27 V
driver #1 (10 k)	1.65 V
#1 + #2	1.34 V
MP2721 (4.7 k)	1.06 V
MP2721 + #2	0.92 V
MP2721 + #1	0.80 V
all three	0.72 V

Loose Notes

Low Frequency IO Pins:
= P0.02 P0.03 P0.09 P0.10 P0.28 P0.29 P0.30 P0.31
= P1.01 P1.02 P1.03 P1.04 P1.05 P1.06 P1.07 P1.10
= P1.11 P1.12 P1.13 P1.14 P1.15

Standard Frequency IO Pins:
= P0.00 P0.01 P0.04 P0.05 P0.06 P0.07 P0.08 P0.11
= P0.12 P0.13 P0.14 P0.15 P0.16 P0.17 P0.18 P0.19
= P0.20 P0.21 P0.22 P0.23 P0.24 P0.25 P0.26 P0.27
= P1.00 P1.08 P1.09

P0.18 is the nRESET Pin. The datasheet is unclear about how reset works

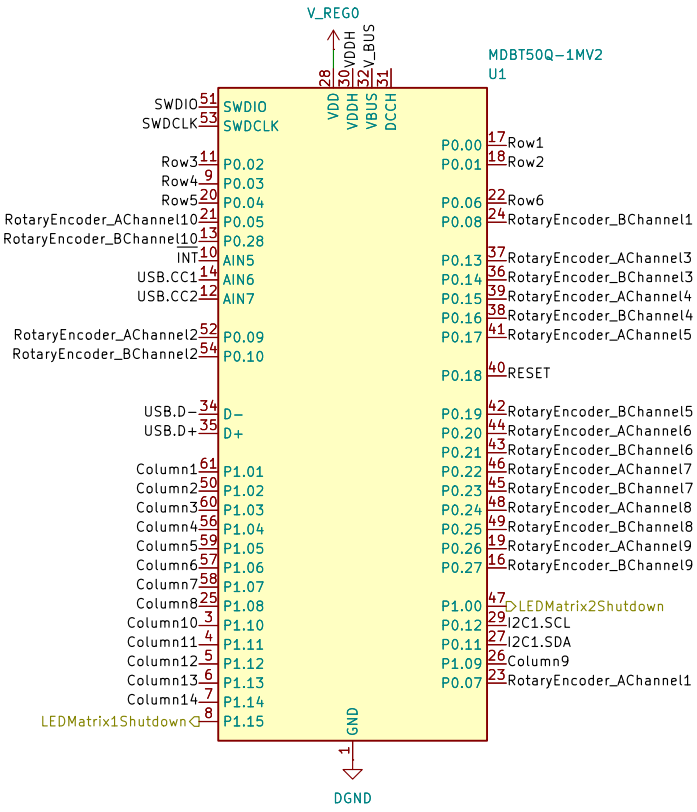
Be careful with the pin assignments as not every pin can handle a high speed signal

Almost any pin can be used for i2c on the nrf52. Sight this source
So I'll use P0.00 and P0.01.
This might be problematic as these pins are used for the secondary clock.

Consider if there enough space for UART debug. No there will not be

Set up resets for the device.
Centralize the decoupling for VDDH

Column[1..14]
Row[1..6]
RotaryEncoder_AChannel[1..10]
RotaryEncoder_BChannel[1..10]
USB{D+, D−, CC1, CC2}



Sheet: /MCU/		
File: controller.kicad_sch		
Title: Slack		
Size: User	Date: 2026-07-25	**Rev: A.0.7**
KiCad E.D.A. 10.0.5		Id: 3/7

Main Matrix

Column[1..14]
Row[1..6]
RotaryEncoder_AChannel[1..10]
RotaryEncoder_BChannel[1..10]

Row1
Row2
Row3
Row4
Row5
Row6

Latch Row

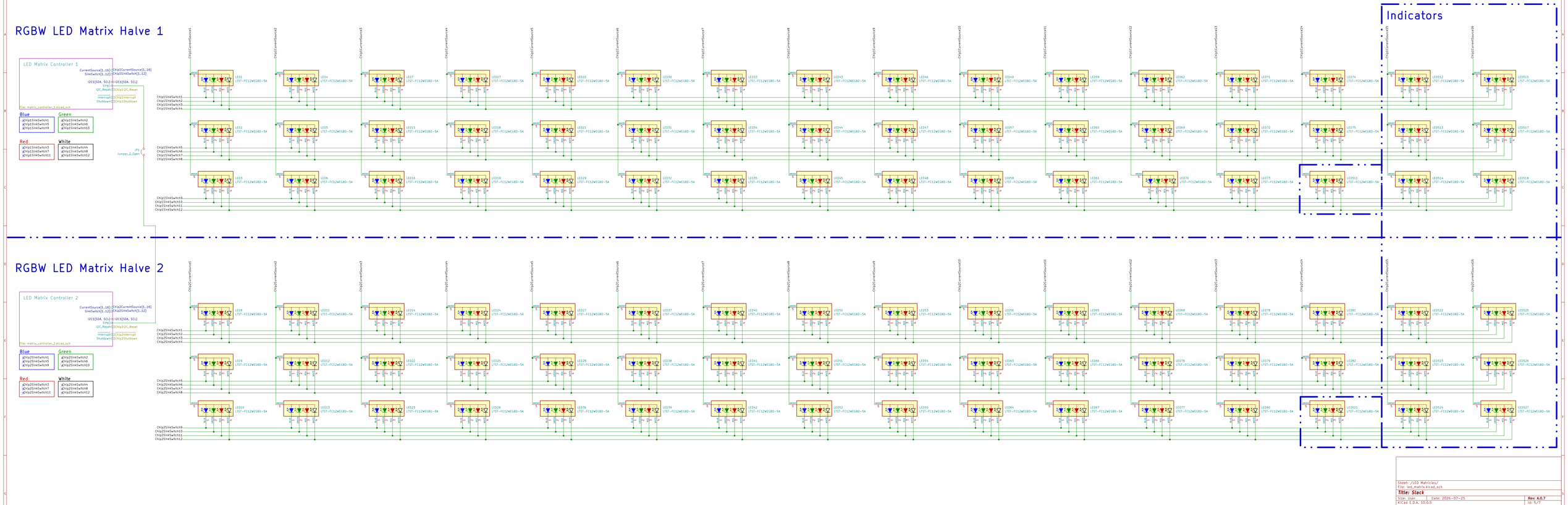
V_REG0
DGND

RE1/SW7 PEC11S-929K-S0015
RE2/SW8 PEC11S-929K-S0015
RE3/SW21 PEC11S-929K-S0015
RE4/SW22 PEC11S-929K-S0015
RE5/SW35 PEC11S-929K-S0015
RE6/SW36 PEC11S-929K-S0015
RE7/SW49 PEC11S-929K-S0015
RE8/SW50 PEC11S-929K-S0015
RE9 EC10E1220505
RE10 EC10E1220505

RotaryEncoder_AChannel1
RotaryEncoder_BChannel1
RotaryEncoder_AChannel2
RotaryEncoder_BChannel2
RotaryEncoder_AChannel3
RotaryEncoder_BChannel3
RotaryEncoder_AChannel4
RotaryEncoder_BChannel4
RotaryEncoder_AChannel5
RotaryEncoder_BChannel5
RotaryEncoder_AChannel6
RotaryEncoder_BChannel6
RotaryEncoder_AChannel7
RotaryEncoder_BChannel7
RotaryEncoder_AChannel8
RotaryEncoder_BChannel8
RotaryEncoder_AChannel9
RotaryEncoder_BChannel9
RotaryEncoder_AChannel10
RotaryEncoder_BChannel10

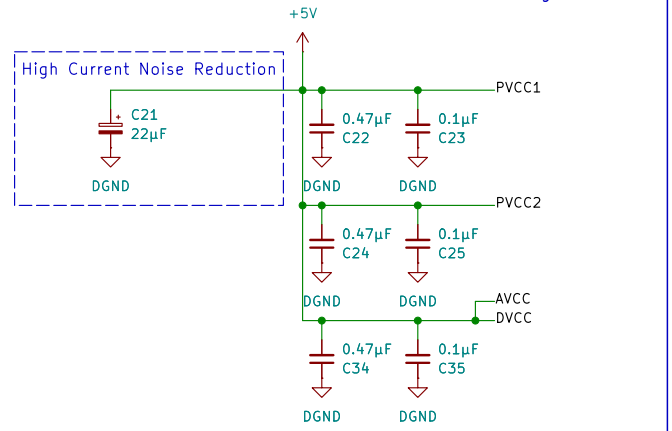
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Size: User	Date: 2026-07-25
KiCad E.D.A. 10.0.5	Id: 4/7

Rev: A.0.7

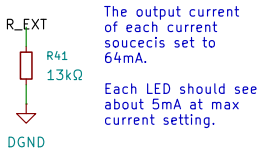

RGBW LED Matrix Halve 1
LED Matrix Controller 1
CurrentSource[1..16]
SinkSwitch[1..12]
I2C1(SDA, SCL)
SYNC
I2C_Reset
Interrupt
Shutdown
Chip1CurrentSource[1..16]
Chip1SinkSwitch[1..12]
I2C1(SDA, SCL)
Chip1I2C_Reset
Chip1Interrupt
Chip1Shutdown
File: matrix_controller_1.kicad_sch
Blue
Chip1SinkSwitch1
Chip1SinkSwitch5
Chip1SinkSwitch9
Green
Chip1SinkSwitch2
Chip1SinkSwitch6
Chip1SinkSwitch10
Red
Chip1SinkSwitch3
Chip1SinkSwitch7
Chip1SinkSwitch11
White
Chip1SinkSwitch4
Chip1SinkSwitch8
Chip1SinkSwitch12
JP1
Jumper_2_Open
Chip1CurrentSource1
Chip1CurrentSource2
Chip1CurrentSource3
Chip1CurrentSource4
Chip1CurrentSource5
Chip1CurrentSource6
Chip1CurrentSource7
Chip1CurrentSource8
Chip1CurrentSource9
Chip1CurrentSource10
Chip1CurrentSource11
Chip1CurrentSource12
Chip1CurrentSource13
Chip1CurrentSource14
Chip1CurrentSource15
Chip1CurrentSource16
LED1 LED2 LED3 LED4 LED5 LED6 LED7 LED15 LED16 LED17 LED18 LED19 LED20 LED21 LED29 LED30 LED31 LED32 LED33 LED34 LED35 LED43 LED44 LED45 LED46 LED47 LED48 LED49 LED57 LED58 LED59 LED60 LED61 LED62 LED70 LED71 LED72 LED73 LED74 LED75 LED511 LED512 LED513 LED514 LED515 LED516 LED517 LED518
LED-RGBW
LTST-FC12WEGBD-5A
Indicators
RGBW LED Matrix Halve 2
LED Matrix Controller 2
CurrentSource[1..16]
SinkSwitch[1..12]
I2C1(SDA, SCL)
SYNC
I2C_Reset
Interrupt
Shutdown
Chip2CurrentSource[1..16]
Chip2SinkSwitch[1..12]
I2C1(SDA, SCL)
Chip2I2C_Reset
Chip2Interrupt
Chip2Shutdown
File: matrix_controller_2.kicad_sch
Blue
Chip2SinkSwitch1
Chip2SinkSwitch5
Chip2SinkSwitch9
Green
Chip2SinkSwitch2
Chip2SinkSwitch6
Chip2SinkSwitch10
Red
Chip2SinkSwitch3
Chip2SinkSwitch7
Chip2SinkSwitch11
White
Chip2SinkSwitch4
Chip2SinkSwitch8
Chip2SinkSwitch12
Chip2CurrentSource1
Chip2CurrentSource2
Chip2CurrentSource3
Chip2CurrentSource4
Chip2CurrentSource5
Chip2CurrentSource6
Chip2CurrentSource7
Chip2CurrentSource8
Chip2CurrentSource9
Chip2CurrentSource10
Chip2CurrentSource11
Chip2CurrentSource12
Chip2CurrentSource13
Chip2CurrentSource14
Chip2CurrentSource15
Chip2CurrentSource16
LED8 LED9 LED10 LED11 LED12 LED13 LED14 LED22 LED23 LED24 LED25 LED26 LED27 LED28 LED36 LED37 LED38 LED39 LED40 LED41 LED42 LED50 LED51 LED52 LED53 LED54 LED55 LED56 LED63 LED64 LED65 LED66 LED67 LED68 LED76 LED77 LED78 LED79 LED80 LED81 LED82 LED519 LED520 LED521 LED522 LED523 LED524 LED525 LED526 LED527
LED-RGBW
LTST-FC12WEGBD-5A
Sheet: /LED Matrices/
File: led_matrix.kicad_sch
Title: Slack
Size: User
Date: 2026-07-25
Rev: A.0.7
KiCad E.D.A. 10.0.5
Id: 5/7

LED Matric Controller 1

Power Config



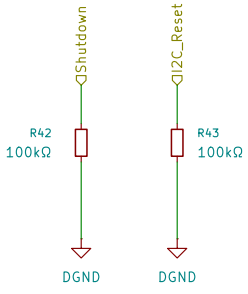
LED max current setting



The output current of each current soucecis set to 64mA.

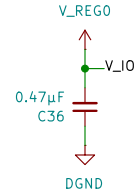
Each LED should see about 5mA at max current setting.

Signal Config

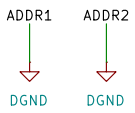


I2C Config

logic reference voltage



I2C address setting



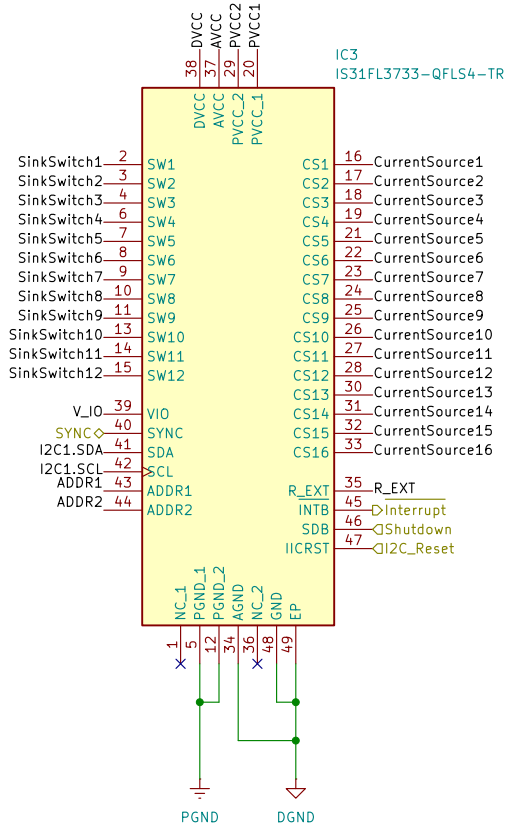
CurrentSource[1..16]
SinkSwitch[1..12]
I2C1{SDA, SCL}
SYNC
I2C_Reset
Shutdown
Interrupt

Blue
- SinkSwitch1
- SinkSwitch5
- SinkSwitch9

Green
- SinkSwitch2
- SinkSwitch6
- SinkSwitch10

Red
- SinkSwitch3
- SinkSwitch7
- SinkSwitch11

White
- SinkSwitch4
- SinkSwitch8
- SinkSwitch12



*is the use of 2 seperate grounds necessary for this?

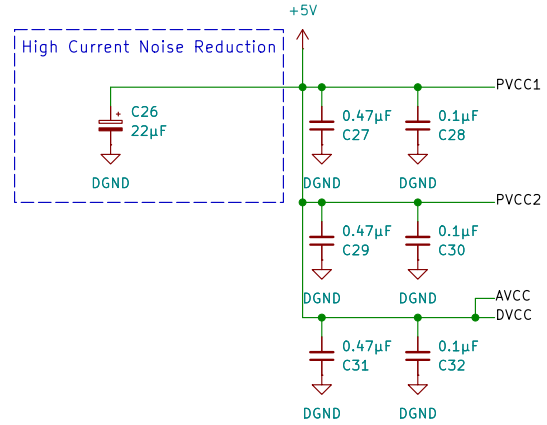
Sheet: /LED Matricies/LED Matrix Controller 1/
File: matric_controller_1.kicad_sch

Title: Slack

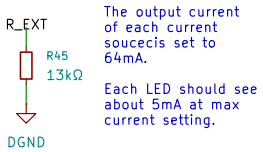
Size: A	Date: 2026-07-25	**Rev: A.0.7**
KiCad E.D.A. 10.0.5		Id: 6/7

LED Matric Controller 1

Power Config



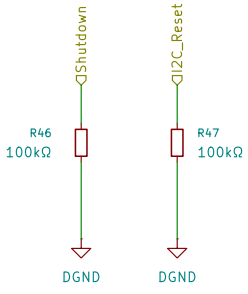
LED max current setting



The output current of each current soucecis set to 64mA.

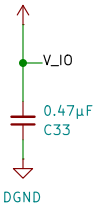
Each LED should see about 5mA at max current setting.

Signal Config



I2C Config

logic reference voltage



I2C address setting



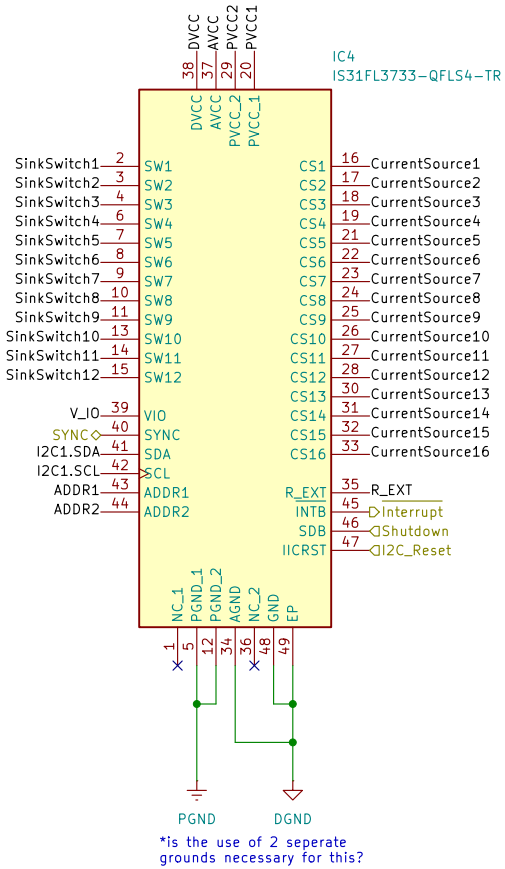
CurrentSource[1..16]
SinkSwitch[1..12]
I2C1{SDA, SCL}
SYNC
I2C_Reset
Shutdown
Interrupt

Blue
- SinkSwitch1
- SinkSwitch5
- SinkSwitch9

Green
- SinkSwitch2
- SinkSwitch6
- SinkSwitch10

Red
- SinkSwitch3
- SinkSwitch7
- SinkSwitch11

White
- SinkSwitch4
- SinkSwitch8
- SinkSwitch12



*is the use of 2 seperate grounds necessary for this?

Sheet: /LED Matricies/LED Matrix Controller 2/
File: matrix_controller_2.kicad_sch

Title: Slack

Size: A	Date: 2026-07-25	**Rev: A.0.7**
KiCad E.D.A. 10.0.5		Id: 7/7